

Statistical modelling of Stocks Returns using MLE and Hypothesis Testing

- **Which distribution best models the returns?**

Based on the KS goodness-of-fit test, the Laplace Distribution was found to be the best model for daily returns over the period. It provided the best fit among the other distributions as it was having the highest p-value among 5 distributions....

- **What are the estimated parameters?**

Normal Distribution (MLE):
Location: 0.0018567742795088894
Scale: 0.014666963830637035

T Distribution (MLE):
Degree of Freedom: 4.486293269631551
Location: 0.0011628190583319545
Scale: 0.011216994438423791

Logistic Distribution (MLE):
Location: 0.0013521624305725787
Scale: 0.007856467457727308

Cauchy Distribution (MLE):
Location: 0.0006000784626970852
Scale: 0.007074336431509499

Laplace Distribution (MLE):
Location: 0.0006340758915526927
Scale: 0.010648207974228465

- **What are the test results and p-values?**

	Distribution	KS Statistic	p-value
0	Laplace	0.039310	0.776
1	Student's T	0.043804	0.675
2	Logistic	0.055729	0.360
3	Cauchy	0.063761	0.221
4	Normal	0.086736	0.050

Normal distribution best fits the return behavior qualitatively according to the KS statistics. The Laplace distribution had the highest p-value in the KS test, saying that it best aligns with the empirical distribution under hypothesis testing...

- **What do the confidence interval outliers indicate?**

Outliers are just returns falling outside the 95% confidence interval for the mean of daily returns under the fitted distribution (here, Laplace distribution)...

These outliers represent statistically unusual return events like:

global events

Macroeconomic news

volatility spikes

Plotting these events in the time series helps us identify when significant deviations from expected behaviour occurred which is useful for risk management and trading alerts based on unusual events...